

Post-Selected Quantum Metrology

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Regardless of the quantity being sensed, quantum sensing requires an interaction between the system and the target. In our NV center system, we sense an external magnetic field, captured by the Hamiltonian $\mathcal{H}_B = \gamma B S_z$, where γ is the gyromagnetic ratio. Through this interaction term and the time evolution $e^{-i\mathcal{H}_B t}$, information about the B -field is imprinted onto the state: $|\psi\rangle \rightarrow e^{-i\mathcal{H}_B t}|\psi\rangle$.

The quantum Fisher information (QFI) quantifies how rapidly a quantum state, represented by a density matrix, changes with respect to the B -field. The faster the change, the greater the sensitivity:

$$F_Q = \text{Tr} \rho L^2.$$

We hypothesize that adding a post-selection step can be advantageous. Intuitively, the B -field information is encoded in the phase, and post-selection lets us discard the unnecessary part of the state into an ancillary qubit (or the other energy level). For a probe coupled to an ancilla,

$$\rho = p_0|0\rangle\langle 0| \otimes \rho_0 + p_1|1\rangle\langle 1| \otimes \rho_1 \longrightarrow \begin{cases} \rho_0 & \text{if the ancilla is measured in } |0\rangle, \text{ with probability } p_0, \\ \rho_1 & \text{if the ancilla is measured in } |1\rangle, \text{ with probability } p_1. \end{cases}$$

Last time I ran numerical simulations to identify which probe state and post-selection filter achieve the maximum QFI. I had tried numerical simulation:

$$\max_{\theta, U, V} F_C^B$$

Recently, I got insight to analyze QFI enhancement.

The state evolves through four stages:

$$|\psi\rangle \xrightarrow{\text{Time evo}} |\psi_0\rangle = e^{-iH_B\tau} |\psi\rangle \xrightarrow{\text{Phase Damping}} \rho = \mathcal{D}(\rho_0) \xrightarrow{\text{Post-selection}} \sigma = \frac{K\rho K^\dagger}{\text{Tr} K\rho K^\dagger}.$$

Without loss of generality, write the pure probe state in the computational basis:

$$|\psi\rangle = \sum_x a_x |x\rangle, \quad \sum_x |a_x|^2 = 1,$$

where x ranges over computational-basis bitstrings like $|00 \cdots 0\rangle$.

Time evolution. Under the detuned magnetic field,

$$|\psi_0\rangle = e^{-iH_B\tau} |\psi\rangle = \sum_x a_x e^{i\theta|x|} |x\rangle,$$

Post-Selection: Phase-Damping Channel

The density matrix before damping is

$$\rho_0 = \sum_{x,y} a_x \bar{a}_y e^{i\theta(|x|-|y|)} |x\rangle \langle y|.$$

After phase damping, off-diagonal elements decay by $\eta^{d(x,y)}$, where $d(x,y) = |x \oplus y|$ is the Hamming distance:

$$\begin{aligned} \rho_0 \rightarrow \rho &= \sum_{x,y} a_x \bar{a}_y e^{i\theta(|x|-|y|)} \eta^{d(x,y)} |x\rangle \langle y| \\ &= U(\theta) D_a C_N D_a^\dagger U^\dagger(\theta), \end{aligned}$$

with $D_a = \text{diag}(a_x)$ and $C_N = (I + \eta X)^{\otimes N}$.

Phase damping kills coherences across different Hamming weights but preserves populations $|a_x|^2$, so ρ has the eigenvalue structure of a *mixed* state on $\text{supp} \rho$.

Post-Selection: Filter Step

A θ -independent Kraus operator K (with $K^\dagger K \preceq I$) is applied to ρ , producing the success branch

$$\sigma = \frac{K\rho K^\dagger}{p_s}, \quad p_s = \text{Tr}[K\rho K^\dagger].$$

The conditional phase QFI

$$F_{\theta,Q}^{\text{ps}} = \text{Tr}(\sigma L_\sigma^2)$$

is the quantity we bound and optimize in the remaining slides.

Post-Selection: Quick Summary

For an N -qubit system, the phase QFI of the post-selected state obeys

$$F_{\theta,Q}^{\text{ps}} \leq N^2 \frac{\eta^2}{1 - \eta^2}, \quad \eta = e^{-\tau_d}, \quad \tau_d = (\tau/T_2)^p.$$

Strategy. Chain the relevant inequalities and analyze when each saturates. From the saturation conditions we can answer:

- Which probe states satisfy equality? What is the corresponding filter basis?
- Is there an entanglement-related gain?
- What is the post-selection success probability?
- Which probe state maximizes that success probability?

For an N -qubit system, the phase QFI of the post-selected state obeys

$$F_{\theta,Q}^{\text{ps}} \leq N^2 \frac{\eta^2}{1 - \eta^2}, \quad \eta = e^{-\tau_d}, \quad \tau_d = (\tau/T_2)^p.$$

- **Which probe states satisfy equality? What is the corresponding filter basis?**
Any $|\psi\rangle = \sum_x a_x |x\rangle$ with $a_x \neq 0$ for all x attains the maximum; the corresponding filter basis is derived below.
- **Is there an entanglement-related gain?**
No — the saturating class is dense and includes product states, so QFI alone shows no entanglement-related gain.
- **What is the post-selection success probability, and which state maximizes it?**
Among saturating probes, the state that maximizes the success probability is symmetric under both qubit exchange and bit flip.

The post-selected density matrix is

$$\sigma = \frac{K\rho K^\dagger}{p_s}, \quad p_s = \text{Tr} K\rho K^\dagger.$$

The QFI is obtained from the symmetric logarithmic derivative (SLD):

$$F_Q = \text{Tr} \sigma L^2, \quad \dot{\sigma} = \frac{1}{2}(\sigma L + L\sigma).$$

Computing the SLD directly is generally hard, so we work with another operator instead.

Post-Selection: Bounding via M

Introduce $M = \sigma^{-1/2} \dot{\sigma} \sigma^{-1/2}$. Substituting the SLD definition,

$$M = \frac{1}{2} (\sigma^{1/2} L \sigma^{-1/2} + \sigma^{-1/2} L \sigma^{1/2}).$$

The key inequality is $\boxed{\text{Tr} \sigma L^2 \leq \text{Tr} \sigma M^2}$.

In the eigenbasis $\sigma = \sum_k p_k |p_k\rangle \langle p_k|$ with $p_k \neq 0$,

$$L_{ab} = \frac{2\dot{\sigma}_{ab}}{p_a + p_b}, \quad M_{ab} = \frac{\dot{\sigma}_{ab}}{\sqrt{p_a p_b}}.$$

$$\text{Tr} \sigma L^2 = \sum_{a,b} \frac{2|\dot{\sigma}_{ab}|^2}{p_a + p_b}, \quad \text{Tr} \sigma M^2 = \sum_{a,b} \frac{|\dot{\sigma}_{ab}|^2 (p_a + p_b)}{2p_a p_b}.$$

Term by term, $m_{ab}/l_{ab} = (p_a + p_b)^2 / (4p_a p_b) \geq 1$, with **saturation** when $p_a = p_b$.

Post-Selection: Lifting to ρ

Here, σ lives in the subspace of ρ selected by K . Accordingly, M lives in that same subspace, and its operator norm is bounded by that of M_ρ :

$$\text{Tr} \sigma M^2 \leq \|M_\rho\|_\infty^2.$$

On the other hand,

$$M_\rho = \rho^{-1/2} \dot{\rho} \rho^{-1/2}, \quad \dot{\rho} = -i[H_B, \rho] = \sum_i -\frac{i}{2} [Z_i, \rho].$$

Writing $M_\rho = \sum_i M_i$ with $M_i = \rho^{-1/2} \left(-\frac{i}{2} [Z_i, \rho]\right) \rho^{-1/2}$,

$$\|M_\rho\|_\infty \leq \sum_i \|M_i\|_\infty.$$

Here, $C_N = C^{\otimes N}$ and $[H_B, D_a] = 0$ force equality for any full-support probe.

Putting it together,

$$F_Q = \text{Tr} \sigma L^2 \leq \text{Tr} \sigma M_\sigma^2 \leq \|M_\rho\|_\infty^2 \leq \left(\sum_i \|M_i\|_\infty \right)^2.$$

When every step saturates simultaneously,

$$F_Q \leq N^2 \|M_i\|_\infty^2.$$

Three saturation conditions:

1. σ is maximally mixed on the filter subspace.
2. The optimal filter is chosen.
3. (Automatically satisfied here — see the previous slide.)

Next, we evaluate $\|M_i\|_\infty$.

To extract the eigenvalues and eigenvectors of M , it helps to recast the problem as a generalized eigenvalue problem:

$$\begin{aligned} M|v\rangle &= \lambda|v\rangle \\ \rho^{-1/2}\dot{\rho}_j\rho^{-1/2}|v\rangle &= \lambda|v\rangle \\ \dot{\rho}_j(\rho^{-1/2}|v\rangle) &= \lambda\rho(\rho^{-1/2}|v\rangle) \\ \dot{\rho}_j|w\rangle &= \lambda\rho|w\rangle, \end{aligned}$$

with $|w\rangle = \rho^{-1/2}|v\rangle$. The last line is the GEP we solve.

Post-Selection: Generalized Eigenvalue Problem

The standard eigenvalue problem is $A\mathbf{x} = \lambda\mathbf{x}$, with eigenvalues from $\det(A - \lambda I) = 0$.

The **generalized eigenvalue problem (GEP)** extends this to $A\mathbf{x} = \lambda B\mathbf{x}$. This should be familiar from, e.g., coupled mechanical systems: the Lagrangian gives

$$M\ddot{\mathbf{x}} + K\mathbf{x} = 0.$$

A harmonic ansatz $\mathbf{x} = \mathbf{x}_0 e^{i\omega t}$ then yields

$$K\mathbf{x} = \omega^2 M\mathbf{x},$$

a GEP whose generalized eigenvalues are the squared normal-mode frequencies.

The relevant block-form is

$$\dot{\rho}_j = \begin{pmatrix} 0 & -i\eta B_j \\ i\eta B_j^\dagger & 0 \end{pmatrix},$$

so the GEP reduces to

$$\begin{pmatrix} 0 & -i\eta B_j \\ i\eta B_j^\dagger & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \lambda \begin{pmatrix} A_j & \eta B_j \\ \eta B_j^\dagger & D_j \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}.$$

Post-Selection: Schur Decomposition

Applying a Schur decomposition with $B_j = A_j^{1/2} W_j D_j^{1/2}$,

$$\begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} 0 & -i\eta W_j \\ i\eta W_j^\dagger & 0 \end{pmatrix} \begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \\ = \lambda \begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} I & \eta W_j \\ \eta W_j^\dagger & I \end{pmatrix} \begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}.$$

Stripping the outer factors ($A_j^{1/2} = \tilde{w}_1, D_j^{1/2} = \tilde{w}_2$),

$$\begin{pmatrix} 0 & -i\eta W_j \\ i\eta W_j^\dagger & 0 \end{pmatrix} \begin{pmatrix} \tilde{w}_1 \\ \tilde{w}_2 \end{pmatrix} = \lambda \begin{pmatrix} I & \eta W_j \\ \eta W_j^\dagger & I \end{pmatrix} \begin{pmatrix} \tilde{w}_1 \\ \tilde{w}_2 \end{pmatrix}.$$

Post-Selection: Reducing via SVD

Take the SVD $W_j = U\Sigma_j V^\dagger$ with $\Sigma_j = \text{diag}(s_{jk})$:

$$\begin{pmatrix} 0 & -i\eta s_{jk} \\ i\eta s_{jk} & 0 \end{pmatrix} \begin{pmatrix} \tilde{u}_{jk1} \\ \tilde{u}_{jk2} \end{pmatrix} = \lambda_{jk} \begin{pmatrix} 1 & \eta s_{jk} \\ \eta s_{jk} & 1 \end{pmatrix} \begin{pmatrix} \tilde{u}_{jk1} \\ \tilde{u}_{jk2} \end{pmatrix}.$$

The determinant condition

$$\det \begin{pmatrix} -\lambda & -\eta s_{jk}(\lambda + i) \\ \eta s_{jk}(-\lambda + i) & -\lambda \end{pmatrix} = 0$$

gives

$$\lambda_{jk} = \pm \sqrt{\frac{\eta^2 s_{jk}^2}{1 - \eta^2 s_{jk}^2}}.$$

Post-Selection: Single-Qubit Bound

Since $0 \leq s_{jk} \leq 1$, the magnitude of λ_{jk} is maximized at $s_{jk} = 1$, giving

$$\|M_j\|_\infty \leq \frac{\eta}{\sqrt{1 - \eta^2}}.$$

This is one more inequality, which must hold for every qubit. Generically $s_{jk} < 1$ for mixed states; however, for any initial state $|\psi_0\rangle = \sum_x a_x |x\rangle$ with $a_x \neq 0$ for all x , phase damping still lets us attain $s_{jk} = 1$. (Precisely, this condition is satisfied when we set-up optimal filter.)

At saturation,

$$\|M\|_\infty = \frac{N\eta}{\sqrt{1 - \eta^2}}, \quad F_Q \leq \|M\|_\infty^2 = \frac{N^2\eta^2}{1 - \eta^2}.$$

Post-Selection: Saturating Probe States

Every pure state $|\psi_0\rangle = \sum_x a_x |x\rangle$ with $a_x \neq 0$ for **all** computational-basis strings x saturates the global bound

$$F_{\theta,Q}^{\text{ps}} = \frac{N^2 \eta^2}{1 - \eta^2}.$$

The saturating class is **open** (the condition $|a_x| > 0$ for all x is open) and **dense** (full-support probes are generic) in the projective Hilbert space $\mathbb{CP}^{2^N-1}$.

As shown above, the phase-damped state factorizes as $\rho = D_a C_N D_a^\dagger$ with $C_N = (I + \eta X)^{\otimes N}$. D_a is invertible iff $a_x \neq 0$ for all x . Return to the GEP:

$$\dot{\rho}_j |w\rangle = \lambda \rho |w\rangle.$$

Then

$$\begin{aligned} -i[H_B, D_a C_N D_a^\dagger] |w\rangle &= \lambda D_a C_N D_a^\dagger |w\rangle \\ -iD_a [H_B, C_N] D_a^\dagger |w\rangle &= \lambda D_a C_N D_a^\dagger |w\rangle, \end{aligned}$$

since H_B and D_a commute. Letting $|\epsilon\rangle = D_a^\dagger |w\rangle$, we obtain another GEP:

$$-i[H_B, C_N] |\epsilon\rangle = \lambda C_N |\epsilon\rangle.$$

Post-Selection: Single-Qubit GEP

With $C_N = (I + \eta X)^{\otimes N} = C^{\otimes N}$ and $H_B = \frac{1}{2} \sum_i Z_i$,

$$-\frac{i}{2} \sum_i [Z_i, C^{\otimes N}] |\epsilon\rangle = \lambda C^{\otimes N} |\epsilon\rangle.$$

For the product ansatz $|\epsilon\rangle = \bigotimes_i |\epsilon_i\rangle$, each factor satisfies a single-qubit GEP:

$$-\frac{i}{2} [Z, C] |\epsilon_i\rangle = \lambda C |\epsilon_i\rangle.$$

Since $-\frac{i}{2} [Z, C] = \eta Y$, this reduces to

$$\begin{pmatrix} -\lambda & -\eta(\lambda + i) \\ -\eta(\lambda - i) & -\lambda \end{pmatrix} |\epsilon_i\rangle = 0,$$

yielding $\lambda_{\pm} = \pm\eta/\sqrt{1-\eta^2}$ and $|\epsilon_i^{\pm}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{\pm i\alpha}|1\rangle)$, where $\cos \alpha = -\eta$, $\sin \alpha = \sqrt{1-\eta^2}$.

Post-Selection: Filter Vectors $|w^\pm\rangle$

Returning to $|w\rangle$:

$$|w^\pm\rangle = (D_a^{-1})^\dagger |\epsilon^\pm\rangle,$$

so

$$\dot{\rho}|w^\pm\rangle = \pm \frac{N\eta}{\sqrt{1-\eta^2}} \rho|w^\pm\rangle.$$

Therefore $M = \rho^{-1/2} \dot{\rho} \rho^{-1/2}$ contains the eigenvalues $\pm N\eta/\sqrt{1-\eta^2}$, giving $\|M\|_\infty = N\eta/\sqrt{1-\eta^2}$ and

$$F_{\theta,Q}^{\text{ps}} = \frac{N^2\eta^2}{1-\eta^2},$$

with filter vectors $|w^\pm\rangle$.

Post-Selection: Filter Basis $|\Psi^\pm\rangle$

Since $|w\rangle = \rho^{-1/2}|v\rangle$, the filter basis (denoted $|\Psi^\pm\rangle$) is

$$|\Psi^\pm\rangle = |v^\pm\rangle = \frac{\rho^{1/2}|w^\pm\rangle}{(1 - \eta^2)^{N/2}}.$$

The two basis vectors are mutually orthogonal:

$$\langle\Psi^+|\Psi^-\rangle = \frac{\langle w^+|\rho|w^-\rangle}{(1 - \eta^2)^N} = 0,$$

inherited from ρ -biorthogonality.

Post-Selection: Matched Kraus Operator

The filter is a rank-2 projector onto $\text{span}\{|\Psi^+\rangle, |\Psi^-\rangle\}$:

$$P_{\pm} = |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|.$$

The matched Kraus operator is

$$K = \sqrt{\frac{p_s}{2}} P_{\pm} \rho^{-1/2}.$$

Applying it to ρ :

$$K\rho K^{\dagger} = \frac{p_s}{2} P_{\pm} \rho^{-1/2} \rho \rho^{-1/2} P_{\pm} = \frac{p_s}{2} P_{\pm}^2 = \frac{p_s}{2} P_{\pm}.$$

Post-Selection: Accepted State

The accepted state is

$$\sigma = \frac{K\rho K^\dagger}{p_s} = \frac{P_\pm}{2} = \frac{|\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|}{2}.$$

That is, the accepted state is the *maximally mixed state on the filter subspace* — exactly the structural requirement for QFI saturation.

This form looks unfamiliar, so we re-express the filter as

$$K = |\Psi^+\rangle\langle w^+| + |\Psi^-\rangle\langle w^-|.$$

It can be implemented as $K = \sqrt{1-\gamma}|00\cdots 0\rangle\langle 00\cdots 0| + |00\cdots 0\rangle\langle 10\cdots 0|$ by sandwiching with appropriate unitaries before and after the post-selection step, **setting one post-selection strength to zero and the others to one.**

The POVM constraint is $K^\dagger K \preceq I$. Explicitly,

$$K^\dagger K = \frac{p_s}{2} \rho^{-1/2} P_\pm \rho^{-1/2} = \frac{p_s}{2(1 - \eta^2)^N} \Xi$$

where

$$\begin{aligned} \Xi &= |w^+\rangle\langle w^+| + |w^-\rangle\langle w^-| = (D_a^{-1})^\dagger \Pi D_a^{-1} \\ \Pi &= |\epsilon^+\rangle^{\otimes N} \langle \epsilon^+|^{\otimes N} + |\epsilon^-\rangle^{\otimes N} \langle \epsilon^-|^{\otimes N}. \end{aligned}$$

And its component form is

$$\Pi_{xy} = \frac{1}{2^{N-1}} \cos(\alpha(|x| - |y|)).$$

The constraint $K^\dagger K \preceq I$ requires $\lambda_{\max}(K^\dagger K) \leq 1$, hence

$$p_s \leq \frac{2(1 - \eta^2)^N}{\lambda_{\max}(\Xi)}.$$

The maximum p_s is obtained when this is tight.

Thank You!

Questions & Discussion